Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
C		2632892

Strictly Com(e)puting: An Analysis of Scoring on Dancing with the Stars

Summary

Dancing with the Stars features weekly eliminations based on a combination of judge evaluations and fan voting, the latter of which is not revealed to the public. The show has also utilized various scoring systems throughout its seasons.

To explore these nuances of scoring, we produce estimated fan votes for each contestant. These estimates are used to analyze contestant characteristics that are predictive of success on the show as well as the impact of different scoring methods on elimination outcomes. To this end, we establish several models, namely **Model 1: Quadratic/Integer Programming Model**, **Model 2: Bayesian Popularity Model**, and **Model 3: Multi-Model Feature Analysis**.

For **Model 1**, we derive bounds on each contestant's possible fan votes through mismatches between the judge scorings and eliminations. We formulate **feasible regions** that contain all possible fan vote distributions and explore how different seasons and controversial contestants influence the constraints. We employ **quadratic programming** to model percentage scoring and **integer programming** to model rank scoring, and set objective values that prevent spikes in fan votes to find deterministic solutions for the finale of each season.

For **Model 2**, we establish that a contestant's fan popularity can be understood as a **latent variable** revealed through observed weekly eliminations and judge evaluations. We employ a Bayesian model using Hamiltonian Monte Carlo implemented in rStan to estimate each contestant's latent fan popularity. The elimination likelihood is modeled using a **multinomial logistic** model. The posterior mean is used to obtain an estimate for the proportion of fan support each active contestant has each week, and to finally calculate estimated fan votes. The results match **93.56%** of the constraints obtained from Model 1, and simulated placements show high **Levenshtein** similarity with the original data.

For **Model 3**, we observe how key characteristics of contestants such as their home state, home country, age during the show, ballroom partner, and industry can affect outcomes such as the contestant's fan votes, average judge score, and their placement in a given season. To do this, we perform **multi-model analysis** via **random forest** and a **multi-layer perceptron** and use MDI and SHAP values to quantify the feature importance of the characteristics. We find that age and home country contributes most to a contestant's success.

Using our fan vote estimates, we simulate the show's different scoring systems and our own proposed dynamically weighted scoring system to analyze discrepancies in weekly eliminations. We find little difference in outcomes between the systems used by the show, but observe that our proposed method mitigates controversy in our case studies. We include a report to the producers outlining our results.

Keywords: Quadratic/integer programming; Bayesian inference; Latent variable estimation; Feature importance

Contents

1 Introduction

1.1 Problem Background

Dancing with the Stars (DWTS) is a popular American reality and competition show. Premiering in 2005, the show has produced 34 seasons and over 500 episodes of primetime television. The show is structured around pairing a celebrity with a professional dancer. Each week, the partners choreograph and perform a dance of varying styles. These dances are scored by professional judges, which is combined with fan voting each week to determine eliminations and the finale placements. Multiple or no eliminations can occur each week, at the discretion of the producers.

The judge scores and fan votes are equally weighted to produce placements for the contestants [1]. Excluding the finale, the placements of the contestants each week is unknown, except for the eliminated participant. Moreover, while judge scores are revealed on the show, fan votes are a closely kept trade secret.

Various scoring methods have been implemented throughout the seasons in reaction to elimination controversies. Initially a rank-based equal weighting approach was used to combine judge scores and fan votes, but from season 3, the show switched to a percentage-based approach. Show switched once again in season 28 to a rank-based approach but with judges having complete control over the final elimination.

1.2 Restatement of the Problem

Given the above background, we aim to accomplish the following tasks:

- **Task 1:** Estimate the amount of fan votes an active contestant receives each week.
- **Task 2:** Study the impact of contestant characteristics/features on their success in the show.
- **Task 3:** Study the different outcomes of rank and percentage-based approaches to combining judge and fan votes.
- **Task 4:** Propose a fairer approach to combining judge and fan votes, with applied case study.

1.3 Literature Review

Inferring popularity or preference is often formulated as a latent variable problem, where a latent variable is a "hidden" quantity that can be inferred from the observed data [2] [3]. Latent variable modeling often involves Bayesian models [4] [5]. This approach is well suited to our setting, where fan voting behavior is unobserved and must be derived from weekly performance outcomes.

1.4 Our Work

To estimate fan votes (**Task 1**) we employ two models: a Quadratic/Integer Programming model and a Bayesian Popularity Model. We cross-validate their results. To determine contestant characteristic importance (**Task 2**), we use multiple machine learning models to predict show success from contestant traits, and then employ feature importance metrics post-hoc. Finally, to compare scoring systems (**Task 3**) we use the given judge scores and our fan votes estimates to simulate weekly eliminations via different scoring mechanisms. We then analyze mismatches with the original data to determine underlying mechanism trends. We use this same simulation to explore our own proposed dynamically weighted scoring system (**Task 4**), and validate that it adequately mitigates controversies. Figure 1 details the flow of this work.

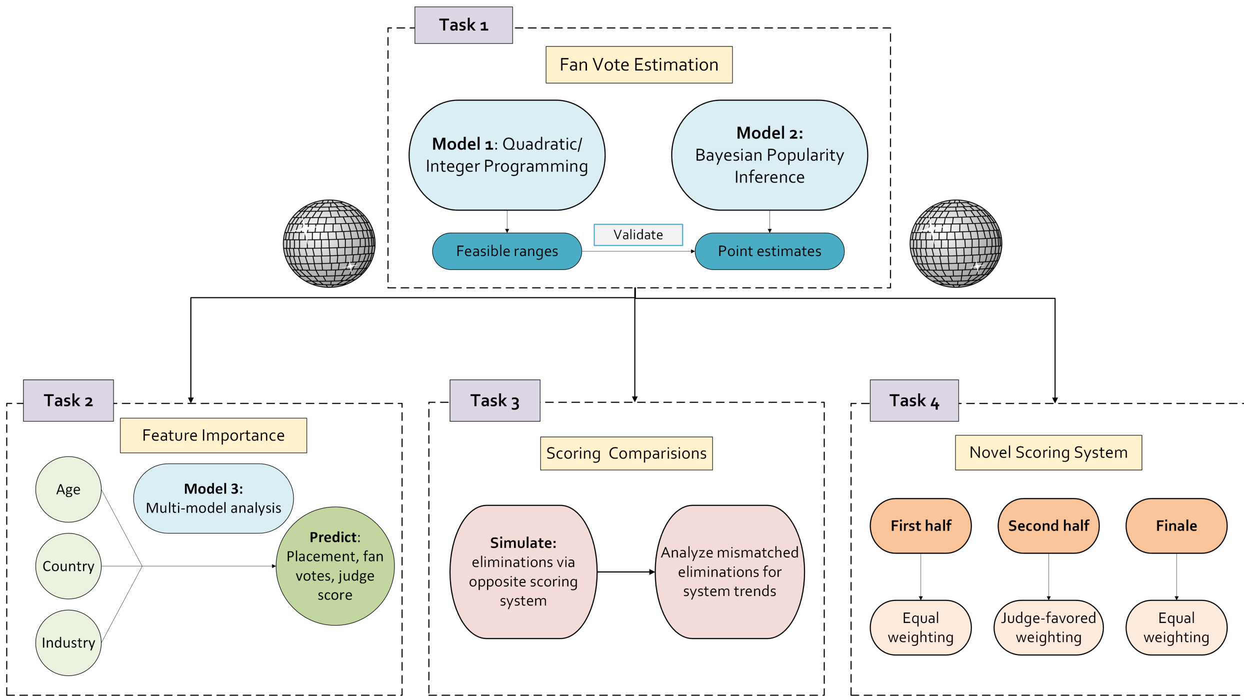


Figure 1: Mindmap for our approach to each task

2 Assumptions and Notations

2.1 Assumptions

Assumption 1: The total fan votes for an episode were $1/2$ the total viewership that episode.

- It is unlikely that all viewers vote, but it is also true that an individual can vote multiple times. Moreover, since our later analysis relies on relative fan votes which is independent from total number, the number of votes can be arbitrary without changing our results. As such, estimating the total votes as one half of total viewership is reasonable.

Assumption 2: No two contestants can receive the same number of votes.

- Millions of fans vote each week, so it is highly unlikely for two contestants to receive the exact same number of votes.

Assumption 3: Two contestants eliminated at once need only have lower scores than the remaining contestants. That is, the rankings between eliminated contestants is inconsequential.

- Regardless of whether one eliminated contestant had a higher score than the other, they will always both be eliminated that week.

Assumption 4: Popularity of each contestant is independent.

- Popularity is not a zero-sum quantity. Multiple contestants can be popular at once. This also provides a neutral starting point for all contestants.

Assumption 5: Popularity of each contestant is stable across the season.

- We are interested in estimating the underlying audience preference throughout the entire show, not shifts in individual contestant popularity week-to-week.

Assumption 6: Weekly outcomes are conditionally independent given contestant popularity and judge scores.

- As voting resets each week, weekly results depend only on what happened that week.

Assumption 7: Judge scores and fan votes contribute equally to weekly outcomes.

- Literature supports this claim for the majority of the seasons [1].

2.2 Notations

We use the following notation in our paper:

Table 1: Notations

Notation	Definition
n	Number of contestants for a given season
W	Number of weeks for a given season
$j_{i,w}$	Judge score for contestant i during week w (w may be implied)
f_i	Latent fan popularity for contestant i
$p_{i,w}$	Fan vote percentage for contestant i during week w (w may be implied)
$r_{i,w}$	Fan vote rank for contestant i during week w (w may be implied)
$s_{i,w}$	Combined score for contestant i during week w
e_w	Elimination or finale outcome in week w
A_w	Active contestants in week w

2.3 Data Collection and Wrangling

Our work relies on the following data sources:

Table 2: Data Source Websites

Data Source	Usage
COMAP 2026 MCM/ICM Problem C	Obtained the given MCM dataset for Problem C, which contained score data for each episode of every season and data on the contestants.
Wikipedia (sourced from LA Times)	Scraped available viewership data for each episode each season, originally sourced from LA Times archives.

Combining and expanding upon these datasets, we use the following data in our work:

Table 3: Data description for by-week dataset

Variable	**Explanation**	**Example**
season	Season of the show	1,2,3, . . .
week	Week of episode for given season	1,2,3, . . .
celebrity_name	Name of celebrity contestant	Mark Cuban, . . .
percent_judge_score	Contestant's percent of total judge points that week	0.13, 0.5, ...
rank_judge_score	Contestant's rank from judge scores that week, ties are handled with competition ranking	1,4,6 ...
winner	Binary whether constestant won	0,1
eliminated	Binary whether contestant was eliminated that week	0,1
ballroom_partner	Name of professional dancer partner	Derek Hough, . . .
celebrity_industry	Celebrity profession category	Athlete, Model, . . .
celebrity_homestate	Celebrity home state (if from U.S.)	Ohio, Maine, . . .
celebrity_homecountry/region	Celebrity home country/region	United States, . . .
celebrity_age_during_season	Age of the celebrity in the season	32, 29, . . .
placement	Final place for the season (1 = best)	1, 2, 3, . . .
ratings	Viewership that episode (in millions)	13.480,...
total_votes	Total votes that episode (in millions)	6.74,...

3 Model Construction

3.1 Model 1: Quadratic/Integer Programming

We aim to obtain a rough estimate to **Task 1** by inferring fan vote ranges based on which contestant(s) got eliminated. Our intuition for obtaining the ranges is as follows. All remaining contestants must have received a higher combined judge/fan vote score than the eliminated contestants. However, if the eliminated people didn't get that many judge votes, this does not tell us much. On the other hand, if the eliminations got a lot of judge votes, we get a lot of information, since this means the remaining contestants must have received a lot of fan votes to have stayed in. In essence, controversial seasons give us more information, as they shrink the possible fan vote bounds for the remaining contestants significantly.

We now produce a bounds-based model for both percentage-based and rank-based voting systems. We initially proceed with percentages, and our notation is as follows. For a given week, consider contestants $1, \dots, n$, and let a subset X denote the eliminations. If X is empty, we can't obtain any information, so suppose X is nonempty. Let their judge's percentages and unknown fan vote percentages that week be $j_1, \dots, j_n$, and $p_1, \dots, p_n$, with $j_i, p_i \in [0, 1]$ and $\sum j_i = \sum p_i = 1$.

Now, for any $i \notin X$ and $x \in X$, we know the remaining contestants must have a higher final score than the eliminated, so we can bound $p_i + j_i > p_x + j_x$. The collection of these bounds formulates our initial feasible region for all p_i in $\mathbb{R}^n$. This region is very large — although there exist some 'unrealistic' solutions that aren't worth considering (such as one contestant receiving 100% of all fan votes), there are arguably far more equally realistic solutions in this region that equally satisfy our data. Hence solving a linear program for a singular optimum solution seems unrepresentative.

So we instead use assumptions to slightly modify this feasible region, and solve the inequalities to obtain rough bounds. Let x' be the contestant for which $j_{x'}$ is highest within X, and assume x' receives the lowest votes among the eliminated (and thus the lowest votes overall) by assumption 3. For any $i \neq x'$, regardless of whether $i \in X$, the previous bound $p_i + j_i > p_{x'} + j_{x'}$ holds. We now have a lower bound for p_i for every single contestant, so we substitute this inequality into $\sum p_i$:

$$1 = \sum_{i=1}^{n} p_i > \sum_{i=1}^{n} (p_{x'} + j_{x'} - j_i) = np_{x'} + nj_{x'} - \sum_{i=1}^{n} j_i,$$

and we obtain an upper bound for $p_{x'}$. Note $p_{x'} \geq 0$ by definition. That is, the eliminated contestant x' can receive zero votes or up to some number of votes, but if they receive more votes than this bound, they feasibly cannot have gotten eliminated. We thus have cleanly bounded the fan votes for x'. We repeat this process, excluding x' to find an upper bound for $p_{x''}$, where $x'' \in X$ is the contestant for which $j_{x''}$ is highest within $X - \{x'\}$. Repeating for all eliminations $x_k \in X$ yields

$$0 \leq p_{x_k} < \frac{1 + \sum_{i=1}^{n-k+1} j_i - (n-k+1) j_{x_k}}{n-k+1}. \tag{1}$$

This represents the range of percentages of votes the elimination x_k can receive while still being eliminated. The process also yields a stronger bound for p_i:

$$p_i > \max_k \left(p_{x_k} + j_{x_k} - j_i \right). \tag{2}$$

Any fan vote percentage distribution that matches our data should fit (1) and (2), so we have achieved our goal for obtaining rough bounds, both for the eliminated and the remaining contesants.

However, the final week is modeled differently as contestants are no longer eliminated but instead fully ranked. We thus have a lot more information to work with, and can give more specific fan vote estimates. Suppose contestants $1, \ldots, n$ are ordered by placements. The inequalities (1) and (2) no longer hold, but our initial bound $p_i + j_i > p_{x'} + j_{x'}$ still holds, now for every pair of contestants instead of just between the remaining/eliminated. This gives a far larger number of constraints that formulate a much tighter feasible region in $\mathbb{R}^n$, where solving for a singular optimum makes sense. We thus formulate a program to solve for an optimum:

$$\begin{aligned}
\text{minimize:} \quad & \sum_{i=1}^{n} p_i^2 \\
\text{subject to:} \quad & \sum_{i=1}^{n} p_i = 1, \\
& p_i > p_k + j_k - j_i, \qquad i < k, \\
& p_i \geq 0, \qquad\qquad\quad\;\; i = 1, \ldots, n.
\end{aligned}$$

Notice this program is quadratic. Since we are now solving for an optimum, we are selective of which one we choose. Solutions that indicate one contestant to have received most of the votes is unrealistic, as most voting distributions are relatively even. Thus our objective function is chosen to prevent spikes in fan votes, resulting in a quadratic program instead of linear. Solving for this objective yields a possible fan vote percentage distribution that matches our data.

The rank scoring system follows the same reasoning but is altered to deal with integer ranks. Seasons 1 and 2 that use the rank system eliminate at most one contestant each week, so X contains at most one element, call $x \in X$. Noting $\sum r_i = n(n+1)/2$ and lower rank numbers indicate higher placement, we obtain the integer versions of bounds (1) and (2):

$$\frac{n+1}{2} - j_x + \frac{j_1 + \cdots + j_n}{n} < r_x \leq n \qquad \text{and} \qquad r_i < r_x + j_x - j_i.$$

For the final round, once again the feasible region is much smaller and we construct a separate model, now with integer programming. The variables are set up to ensure $\{r_1, \ldots, r_n\} = \{1, \ldots, n\}$ by using binary decision variables $b_{i,k} \in \{0, 1\}$ where $i, k = 1, \ldots, n$. Hence we have

$$\begin{aligned}
\text{minimize:} \quad & 0 \\
\text{subject to:} \quad & \sum_{i=1}^{n} b_{i,k} = 1, && k = 1, \ldots, n \\
& \sum_{k=1}^{n} b_{i,k} = 1, && i = 1, \ldots, n \\
& \sum_{k=1}^{n} k b_{\ell,k} < \sum_{k=1}^{n} k b_{m,k} + j_m - j_\ell, && \ell < m.
\end{aligned}$$

Notice our objective is now constant. Our feasible region is very tight and contains very minimal lattice points. We also no longer have unrealistic solutions such as one contestant hoarding all votes, since we now use fan vote rankings and not percentages. Thus any viable solution is equally satisfactory, and we can set a constant objective. Solving for this objective yields a possible fan vote ranking that matches our data.

3.2 Model 2: Bayesian Popularity Inference

We now aim to obtain more concrete fan vote estimates for **Task 1**. To eventually find total fan votes, we first find the latent fan popularity of each contestant throughout the show. Probabilistically, we are trying to find the likely values of fan popularity (F) given the observed weekly outcomes (E) and judge scores (J):

$$P(F|E, J). \tag{3}$$

To uncover this distribution, we employ a Bayesian framework:

$$P(F|E, J) \propto P(F)P(E, J|F). \tag{4}$$

Let

$$F = \{f_i : i \in \{1, 2, ..., n\}\}$$

be the set of latent popularity f_i for all contestants. We assume that this popularity is stable over the course of the season (Assumption 5). In other words, instead of analyzing week-to-week changes in popularity, we are rather looking at the season as a whole and exploring relative contestant popularity across the entire season.

Next, let

$$E = \{e_w,\ w \in \{1, 2, \cdots, W\}\}.$$

be the set of weekly outcomes e_w which is either the eliminated contestant or, if it is the finale week, the show winner.

Finally, let

$$J_w = \{j_{a,w},\ a \in A_w\}$$

and

$$J = \{J_w,\ w \in \{1, 2, \cdots, W\}\}.$$

where $j_{a,w}$ is the judge score for active contestant a in week w.

We know that the weekly outcome depends on a combination of judge scores and fan votes. To reflect this in our model setup, we can rearrange the likelihood in Equation 4 using the chain rule for joint probabilities:

$$P(E, J|F) = P(E|J, F)P(J|F).$$

Substituting into Equation 4 yields

$$P(F|E, J) \propto P(F)P(E|J, F)P(J|F).$$

We then recognize that judge scores do not depend on fan votes, meaning $P(J|F) = P(J)$. Since $P(J)$ does not depend on F, it can be absorbed into the proportionality constant. So,

$$P(F|E, J) \propto P(F)P(E|J, F).$$

Finally, we know that weekly outcomes are determined by an equal weighting of judge and fan evaluation. So we model the combined effect of the judge scores and fan popularity of contestant i during week w through the latent total score

$$s_{i,w} = 0.5j_{i,w} + 0.5f_i.$$

So, our final Bayesian formulation is

$$P(F|E, J) \propto P(F)P(E|S).$$

where

$$S_w = \{s_{a,w},\ a \in A_w\}$$

and

$$S = \{S_w,\ w \in W\}.$$

To simplify our model [6], we assume that contestants' popularities are independent (Assumption 4). Thus,

$$P(F) = P(f_1, f_2, ..., f_n) = \prod_{i=1}^{n} P(f_i).$$

We also assume that weekly outcomes are conditionally independent (Assumption 6). So

$$P(E|S) = P(e_1, e_2, ..., e_W|S_1, S_2, ..., S_W) = \prod_{w=1}^{W} P(e_w|S_w).$$

With these assumptions, the posterior distribution can be represented as

$$P(F|E, J) \propto \prod_{i=1}^{n} P(f_i) \prod_{w=1}^{W} P(e_w|S_w).$$

We now set out priors. We have little information about the contestant's popularity before the start of the season. Thus, we use a weakly informative normal prior, as is common in Bayesian modeling [5]. We set

$$f_i \sim N(0, 1).$$

We now formulate our likelihood. Note that for a single week, there are multiple possible outcomes. We thus model weekly outcomes with a multinomial logit approach [6]. The probability that contestant i is eliminated on a non-finale week w is then

$$P(e_w = i|S_w) = \frac{\exp -s_{i,w}}{\sum_{j \in A_w} \exp -s_{j,w}}.$$

For the finale week ℓ, the probability that contestant i wins is

$$P(e_\ell = i|s_{i,\ell}) = \frac{\exp s_{i,\ell}}{\sum_{j \in A_\ell} \exp s_{j,\ell}}.$$

We use $-s_i$ for elimination weeks as lower scores signal elimination, and conversely s_i is used for the likelihood of winning.

We estimate the posterior distribution with rStan which uses Hamiltonian Monte Carlo. The log-likelihood is manually specified in the model, and thus posterior draws provide estimates of the latent fan popularity f_i for each contestant.

Finally, we are interested in getting an estimate of the percentage of fan votes each contestant receives each week. To do this, we exponentiate each posterior draw f_i and normalize across the set of active contestants to estimate the proportion of total fan support each active contestant i receives during week w:

$$p_{i,w} = \frac{\exp f_i}{\sum_{j \in A_w} \exp f_j}.$$

The mean of $p_{i,w}$ over all posterior draws is used as our estimate for the expected percentage of fan votes for a given contestant during the weeks they participated. Using these percentages and our pre-computed estimated total fan votes for each episode, we can obtain the estimated amount of fan votes each contestant acquired each week they were active.

4 Model Results

4.1 Quadratic/Integer Programming Model Results

The programming model yields rough bounds for the fan vote distribution derived from our feasible region. Eliminated contestants have a minimum of 0 votes and some maximum percentage of votes, beyond which they can no longer be eliminated. The remaining contestants do not have a maximum vote cap; any votes they receive only help them remain and eliminate the others. Instead, they have a minimum vote percentage based on how many votes the eliminated contestants received. If the eliminated receive few votes, the remaining contestants need few votes as well, whereas if the eliminated receive many votes, so do the remaining contestants. Our model thus produces two sets of bounds: a 'best-case scenario' and a 'worst-case scenario' for remaining contestants where the eliminated votes are lowest and highest respectively.

For Model 1, we explore Seasons 2 and 27 in detail below as they are both controversial seasons highlighted by the prompt, and serve as motivations for this investigation. We note that most other seasons (including those specifically highlighted by the prompt) also show similar trends.

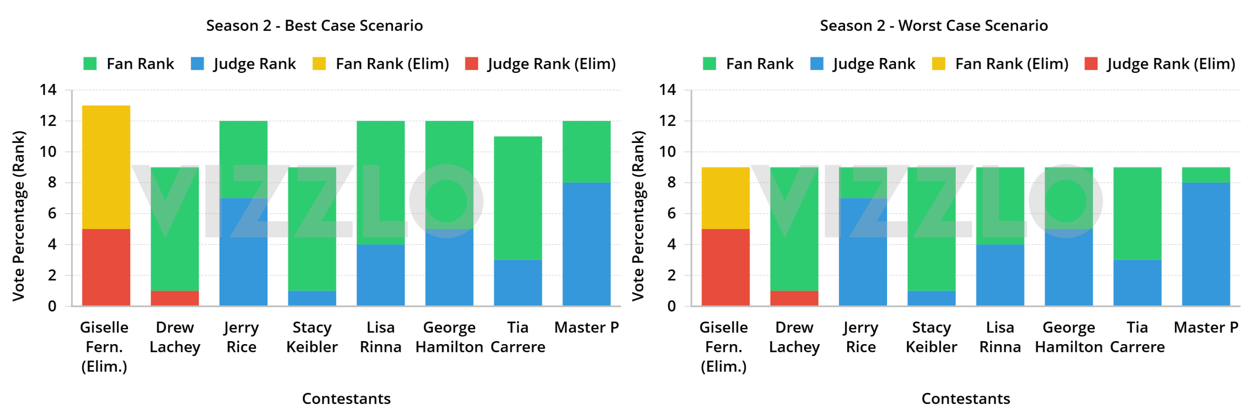


Figure 2: Results for Season 2, Week 3. The lower the rank, the better.

Observe the worst-case and best-case scenarios for Season 2. In week 3 (Figure 2), the maximum fan votes Giselle can receive is rank 5, otherwise Master P with the lowest judge rank would be considered for elimination. Jerry Rice, one of the controversial contestants, also needs at least the second most fan votes in order to avoid elimination. Drew Lachey, with the highest judge rank, can place last by fan votes and still be safe.

Table 4: Results for Season 2, Week 8 (Final 3 Contestants)

Contestant	Judge Rank	Ex. Fan Rank	Rank Sum	Placement
Drew Lachey	1	2	3	First
Jerry Rice	3	1	4	Second
Stacy Keibler	2	3	5	Third

For the finale, our model predicts that Jerry Rice must have been ranked first by fans in order to have placed second overall (Table 4). This explains the controversy around Jerry Rice being a finalist in Season 2.

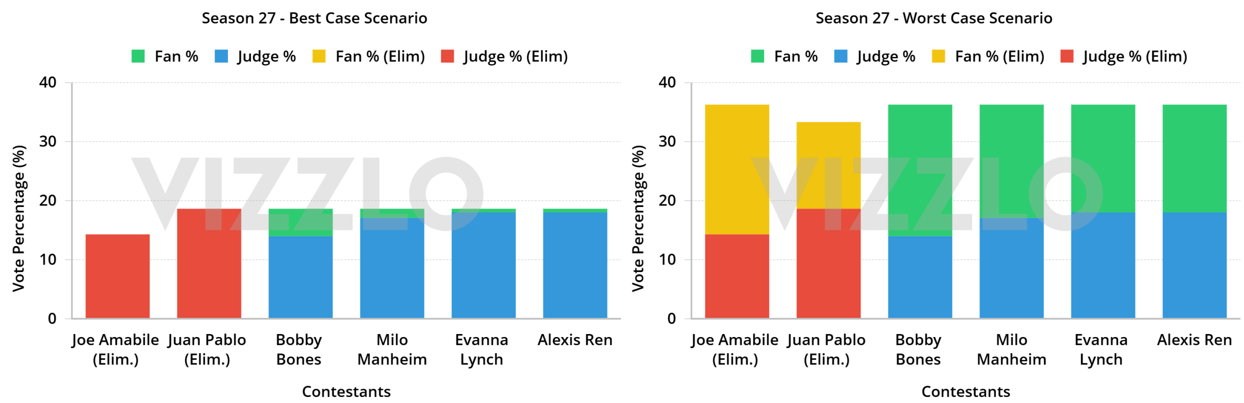


Figure 3: Results for Season 27, Week 8. The higher the %, the better.

Season 27 introduces controversial contestant Bobby Bones. In week 8 (Figure 3), Bobby received the lowest judge scores, so he must receive at least 4.66% of fan votes in the best case scenario, or 22.3% of fan votes in the worst case scenario to avoid elimination.

Table 5: Results for Season 27, Week 9 (Final 4 Contestants)

Contestant	Judge %	Ex. Fan %	Sum %	Placement
Bobby Bones	24.38 %	26.73 %	51.11 %	First
Milo Manheim	25.97 %	24.13 %	50.1 %	Second
Evanna Lynch	25.97 %	24.13 %	50.1 %	Third
Alexis Ren	23.38 %	25.00 %	48.38 %	Fourth

The final round for Season 27 shows Bobby Bones to receive low judge scores, so by our model, Bobby Bones must have received the most fan votes to place first (Table 5). Similarly, Alexis Ren must receive the second most fan votes to ensure the final placements are adhered to.

4.2 Bayesian Popularity Model Results

The Bayesian Popularity Model is employed on a season level. In the interest of time, we look at a subset of seasons: 1, 2, 4, 5, 11, 20, 27, 30, 34. We gave preference to the seasons highlighted in the problem statement as controversial, and gave each type of scoring approach a presence in our sample.

We extract the following data from a given season to pass to Stan:

Notation	Definition
n	Number of contestants in season
W	Number of weeks in season
A_w	Number of active contestants in week w
J	Vector of judge rank/percentages for all weeks
id	Mapping from each observation to a contestant
e_w	Index of the eliminated (or winning) contestant in week w

For seasons 1 and 2, J is the judge ranking for each contestant, and otherwise J is the percentage of judge scores for each contestant. In the Stan model code, the log-likelihood contribution is accumulated manually using `target +=` to account for both elimination and winning weeks:

Algorithm 1 Stan Model Code

for $i \in n$ **do** $F[i] \sim N(0, 1)$
end for

for $w \in W$ **do**
    for $i \in A_w$ **do** $S[i] = 0.5 * J[i, t] + 0.5 * F[i]$
    end for

    if $w < W$ **then** eliminated $\sim$ MultinomialLogit($-S$)
    end if

    if $w = W$ **then** winner $\sim$ MultinomialLogit(S)
    end if
    if w = W **then** $p_{i,w} = \frac{f_i}{\sum_{j \in A_w} f_i}$
    end if
end for

We run our MCMC simulation on 4 chains of length 2000 each. For all seasons, the simulations converged as evidenced by traceplots (Figure 4) and $\hat{R} \approx 1$ for all F_i.

Finally, posterior draws of F_i for active contestants are exponentiated and normalized to compute weekly fan support proportions. We combine with our previously estimated weekly total fan votes to obtain specific estimates for each contestant's weekly fan votes.

We use Season 27 to analyze the the fan vote estimate results. Similar trends are observed across the other computed seasons. Figure 5 illustrates the trajectory of contestant fan votes during Season 27 in comparison to the trajectory of judge scores. Notably, the fan votes fluctuate week-to-week due to changing viewership, but stay consistent relative to one another. This is due to our assumption of stable popularity (Assumption 5). We also see instances where higher popularity does not always align with being safe from elimination, such as contestant 9 eliminated in week 7 and contestant 12 eliminated in week 8. These can be explained by the lower judge scores shown in the right image.

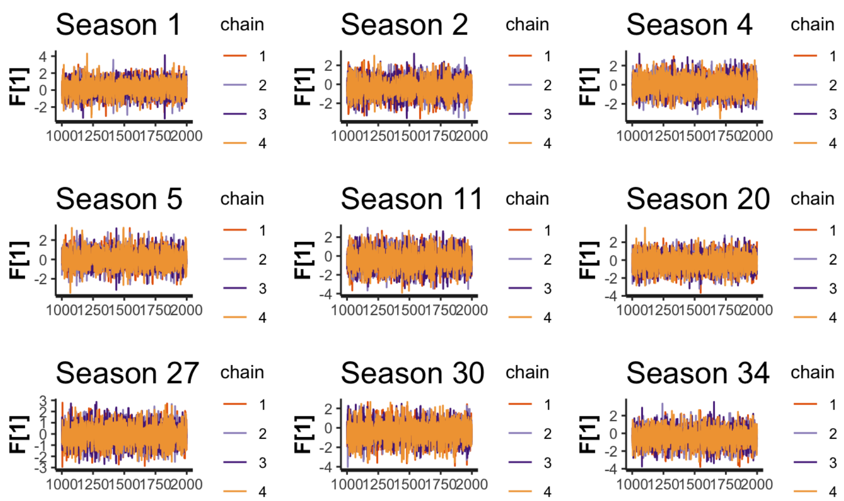


Figure 4: Traceplots for the MCMC of each estimated season

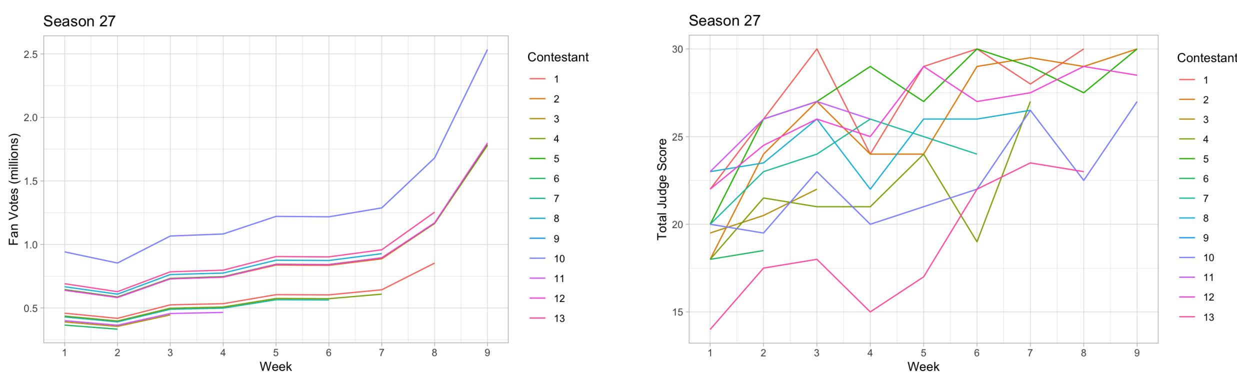


Figure 5: Trajectory of fan votes (left) and judge scores (right) across weeks of Season 27

In fact, contestant 12 had consistently low judge scores, but a high popularity that allowed them to survive for several weeks. This contestant was Alexis Ren, who started a relationship with her dance partner, Alan Bersten, during the season [8]. This romantic subplot likely explains and validates her estimated high popularity.

4.3 Validating the Bayesian Popularity Inference Model

The fan vote distribution predictions from our Bayesian popularity inference model are cross-referenced with our quadratic/integer programming model to see whether the predictions lie in the feasible regions, or equivalently, lie within all the constraints. For all relevant seasons, we have a total of 388 constraints, of which our predictions match 362, yielding an overall match rate of 93.56%, with seasonal match rates varying from 86.66% to 98.48%.

The fan vote distribution predictions are also used to simulate weekly eliminations, which are combined into simulated placements for all relevant seasons. These placements are treated as sequences and are compared to the original placements in the data via the Levenshtein similarity metric. The high similarity values indicate small edit distances overall between the two types of sequences for all considered seasons. Both metrics validate that our predictions match the majority of the data provided (Figure 6).

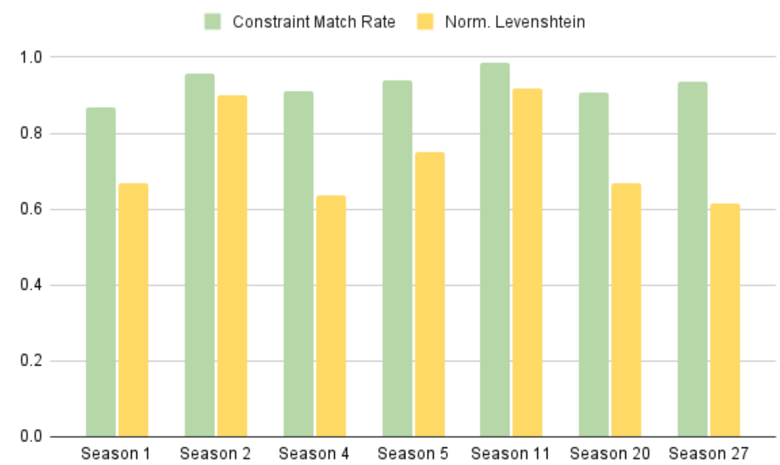


Figure 6: Fit metrics between Bayesian model predictions and season data

We thus complete Task 1: We estimate fan vote percentages through a Bayesian popularity model, and combine it with viewership data to obtain fan vote estimates for each contestant each season. We verify that these estimates match 93.56% of all constraints computed in our quadratic/integer programming model. We also use these estimates to simulate contestant placements, and observe little difference with the original placement data for various seasons. Both validation metrics are displayed in Figure 6.

5 Further Analysis

With our estimated fan votes, we now have a complete understanding of the components that decide a contestant's weekly outcome. We aim to explore how specific contestant characteristics and different scoring systems impact elimination outcomes.

5.1 Model 3: Multi-Model Feature Analysis

We approach **Task 2** by investigating each contestant's dancing partner, industry, home state, home country/region and age, and modeling how each feature influences the contestant placement, judge score averages, and fan percentage estimates derived from our Bayesian popularity model. Note the usage of judge score averages instead of percentages, since percentages inflate as other contestants get eliminated, whereas averages reflect the contestant's true performance.

Initially, we attempted to measure the influence of features using Multiple Linear Regression (MLR). However, the model failed to indicate a significant influence of any features on measures of success. This is because the model failed to capture the variance in outcomes ($R^2 < [X]$). This indicated that the features don't interact linearly and that we needed to use nonlinear feature analysis models to obtain significant results.

We thus use two non-linear data analysis models to extrapolate results. We first apply random forests on all possible feature combinations and use Mean Decrease of Impurity (MDI) to quantify feature importance. We find that age is the top feature for predicting outcomes with an MDI of 59%. In addition, the most successful industries include comedians, singers/rappers, and athletes, which are the most common industries on the show. We reason their frequent appearances attract larger audiences, which increases fan votes for those contestants. We also observe the most successful ballroom partner to be Derek Hough, with 6 wins across 17 seasons and the most perfect judge scores in the show's history.

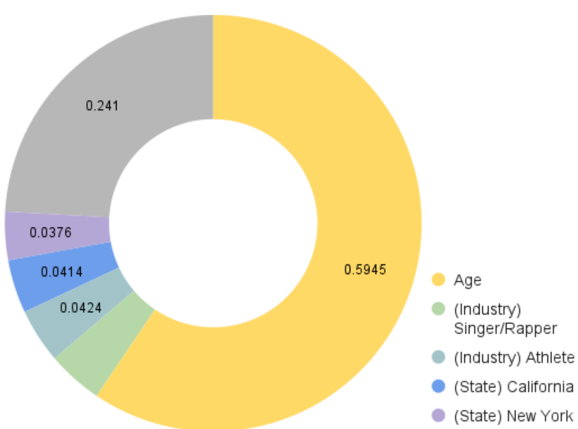
We also employ a multi-layer perceptron (MLP) model with the purpose of modeling non-linear relationships, which we suspect is closer to the data's nature. Since MLP is a black box of sorts we use Shapley Additive Explanations or SHAP values to measure feature importance. Implementing this, we again find age to be the foremost feature. Additionally, whether a contestant's home country was the US ranked second in importance, which we attribute to the vast majority of viewers and contestants of DWTS being American. Contestants being comedians also weighs heavily. We reason that aside from basic popularity, people in such industries have experience quickly adapting to various situations, skills that would provide definite advantages in a show like DWTS.

Age, being the most consistent feature of success, can be explained with a couple of key reasons. Younger contestants tend to be more successful over the run of the show and most given seasons since younger contestants have the stamina to keep up with the demands of the show while still performing at a high level. DWTS demands that stars learn and perfect multiple dances from multiple styles in a week, for potentially up to 3 months, which can be taxing on the body for even the professional dancers, and as such, is also demanding on the stars. Therefore, younger contestants tend to have the stamina needed to meet these demands weekly.

Table 6: Top features for random forest by MDI (left) and MLP by SHAP (right)

Model	Feature	MDI
RF	Age	0.5945
RF	(Industry) Singer/Rapper	0.0431
RF	(Industry) Athlete	0.0424
RF	(State) California	0.0414
RF	(State) New York	0.0376

Model	Feature	SHAP
MLP	Age	0.8602
MLP	(Country) International	0.3993
MLP	(State) California	0.3195
MLP	(Industry) Comedian	0.2431
MLP	(Partner) Derek Hough	0.1879



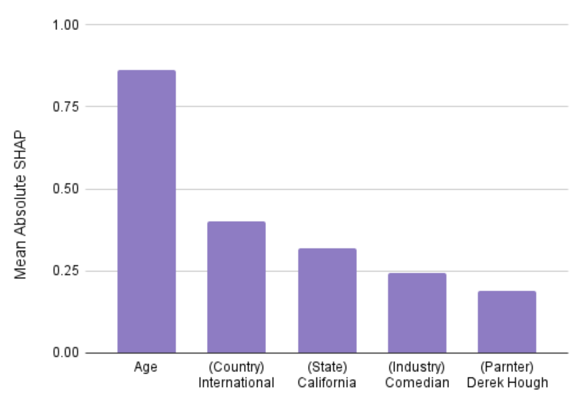


Figure 7: Feature importance for random forest with MDI (left) and MLP with SHAP (right)

Although we analyzed the features against three different metrics, the importance rankings remained consistent across all three, which led us to present the aggregated importance of the characteristics in Table 6 and Figure 7.

We thus complete Task 2: We employ multiple linear regression, random forests, and multi-layer perceptron models to quantify the impact of each contestant's key features on their placement, judge score averages, and fan percentage estimates obtained from our Bayesian popularity model. All models agree that celebrity age, industry, and country are the top predictors of performance. Some models also indicate the pro-dancer partners and their overall performance to strongly impact contestant performances.

5.2 Comparing Scoring Systems

To address **Task 3**, we simulate multiple seasons with the opposite scoring mechanisms — seasons 1,2 and 28-34 will use the percentage-based system and seasons 3-27 will use the rank-based system. Identical to how we validated our Bayesian model results via simulation, we use judge scores from the data and fan vote estimates from our Bayesian popularity model to simulate the eliminated contestant for each given week, and compare with the original eliminations to observe mismatches.

The simulations reveal that the majority of eliminations were consistent via either scoring mechanism. For 96% of all mismatches, we observe that the predicted eliminated contestant had a lower fan score and a higher judge score than the actual eliminated contestant. As this holds across either scoring mechanism, we assume that this error has to do with uncertainty in our fan vote estimates. We conclude that the scoring systems themselves do not inherently differ in favoring either score.

We thus complete Task 3: Using fan vote estimates obtained from our Bayesian popularity model and the judge scores from the data, we simulate weekly outcomes for all seasons with the scoring method other than the ones originally employed. We find little evidence of significant differences in the elimination outcomes and judge/fan favoritism between the two scoring mechanisms, and attribute any observed variance to uncertainty in our fan estimates.

5.3 Proposed Dynamically Weighted Scoring System

Due to the reliance on fan voting to determine outcomes, DWTS has undergone several elimination controversies. For instance, in season 2 Jerry Rice was the runner up but received the lowest judges scores for several weeks. Similarly, in season 4 Billy Ray Cyrus placed 5th despite several weeks of last place judge scores. While clearly these contestants were popular to have survived as long as they did, fans still found their duration on a show fundamentally about dancing distasteful.

It is notable that such controversies have occurred across all the elimination methods that production has employed. As such, to target **Task 4**, we propose a new three-phase dynamically weighted scoring system that can help mitigate the prolongation of contestants who do not meet the dancing standards of the show, while still honoring the viewers' desire to keep certain stars.

We propose a scoring system that does not use fixed scoring weights for judge and fan evaluation throughout the entire season (Figure 8). Equal weighting is used for every week at or before the halfway mark as the fairest way to incorporate both fan and judge preferences. It does not overly penalize contestants unfamiliar with dancing, or unknown to the public. Additionally, it also avoids fan favorites leaving the season early, which would harm viewership and ratings.

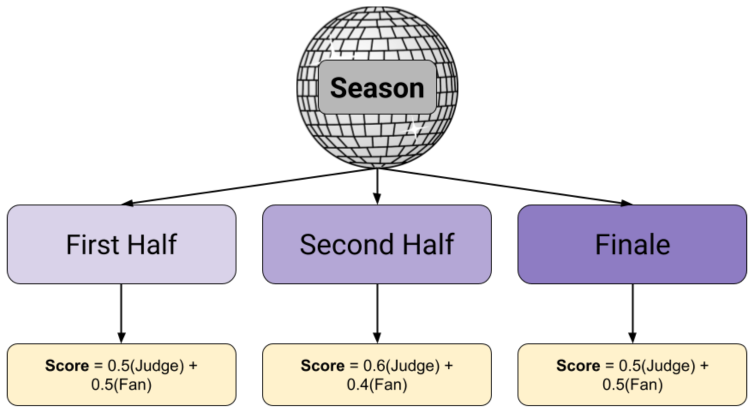


Figure 8: Mindmap of our dynamically weighted scoring system

For every week after the halfway mark (excluding the finale), we change the weighting to favor the judges scores. We first tested a slight increase in judge weighting (60% vs 40%):

$$s_{i,w} = 0.6(j_{i,w}) + 0.4(f_{i,w}).$$

Our case study yielded promising results in eliminating controversy, so in the interest of time we continued with this weighting to avoid undermining fan vote even more. Further sensitivity analysis could be done to find an ideal weighting between judge and fan evaluations.

At this point in the show, it can become grating for audiences to see poor dancers continue to advance without improvement, as evidenced by the highlighted controversies. Thus, we now prioritize dancing ability to weed out contestants undeserving of remaining on the show.

For the finale, we revert back to an equal weighting system. Due to the prior weighting scheme, the finale is now likely to only include the better dancers on the show, so we can reduce the judge score weighting. It is also preferable to allow fan votes to matter more in the finale as fans value say in the final winner.

For a case study, we implemented our method for seasons 2 and 4 using the weekly observed judge scores and estimated fan votes. We found that our method helped mitigate controversy by eliminating the controversial contestant earlier, as desired (Table 7).

Table 7: Placement of controversial contestants under equally vs dynamically weighted scoring

Contestant	Season	Equally Weighted Scoring	Dynamically Weighted Scoring
Jerry Rice	Season 2	Runner up	Eliminated week 7
Billy Ray Cyrus	Season 4	Eliminated week 8	Eliminated week 7

We thus complete Task 4: We propose a new three-phase dynamic scoring system. Our model initially weights fan and judge preferences equally to keep fan favorites and allow for growth, before changing the weights to favor judge scores halfway through to encourage skill and improvement. We switch back to equal weighting for the finale, as fans value having a say in the final winner. Our method adequately balances judge and fan votes, honoring the viewers' desires to keep certain stars while successfully mitigating the prolongation of contestants who do not meet dancing standards, and thus preserves the integrity of the show.

6 Strengths and Weaknesses

6.1 Strengths

- We produce two distinct models to estimate fan votes, and the models validate each others' results. This increases credibility in our estimates.
- Both estimation models are highly adaptable to any given data, as no external information was used for their construction. This also applies to data differences between seasons, where our models show good fitness for most season data regardless of variances in data values.
- Both models also show considerable performance with minimal input data, as evidenced by its success modeling the data provided.
- At its core, Model 1 derives deterministic bounds in data points and is thus conceptually simple and interpretable. Model 2 is also intuitive for this setting as it allows for the information from each week's observed outcomes to be combined to update posterior beliefs.
- Model 2 is computationally inexpensive, involving a single latent parameter per contestant and a closed-form likelihood, which allows for efficient inference via Hamiltonian Monte Carlo.
- For Model 3, the random forest and multi-layer perceptron are good predictors of outcome as they effectively capture the non-linear relations between features and outcomes. The results from each also agree with and thus validate each other.

6.2 Weaknesses

- The accuracy of the bounds of Model 1 relies heavily on the provided data containing numerous mismatches between judge scores and eliminations. For weeks with no eliminations, or with eliminations that align with the judge's scoring, we obtain minimal information and the derived bounds are extremely loose and only prevent edge cases from occurring.
- Model 1 cannot produce specific estimates for fan voting, and the feasible regions formulated for all weeks beside the final week are large, where solving for an optimum become unrepresentative of all the possibilities. An extension of Model 1 could be to tighten the deterministic bounds stochastically, finding appropriate cuts that hold for the majority of possible fan vote distributions.
- Both Model 1 and Model 2 currently assume outcomes week-to-week are independent; an extension for both could incorporate previous week data to carry-over and supplement information when necessary.

- We use the simplifying assumption that contestants' relative fan appeal is consistent over the season for Model 2. While our model succeeds in estimating overall relative fan popularity, we miss changes in a given contestant's popularity as the audience's opinion shifts throughout the show. An extension of Model 2 could predict weekly relative fan popularity by recursively conditioning on the prior weeks' popularity. In our attempts, this method had serious convergence issues, and we found that the simpler Bayesian approach was validated by Model 1 in any case.

7 Conclusion

To estimate fan votes for each contestant for the weeks they competed, we first employed a quadratic/integer programming model to obtain feasible regions for the fan votes. We then obtained point estimates for fan vote proportions through a Bayesian popularity model, which was combined with viewership data to obtain the quantity of votes. We verified that these estimates match 93.56% of all constraints computed in our quadratic/integer programming model. We also observed little difference in placement data when simulating contestant placements with our vote estimates.

With our fan vote estimates, we investigated whether certain contestant traits, such as age, industry, and ballroom partner, are predictive of contestant success. We employed several different models, namely multi-linear regression, random forest, and a multi-layer perceptron, to quantify the importance of these characteristics. We found that age is the most important feature across the models. This is followed by a celebrity being international, celebrity's industry, and a celebrity's country being the top predictors of performance. Having specific ballroom partners who have won a lot, such as Derek Hough, is also a decent predictor across the models.

We also explored discrepancies in placements via different scoring mechanisms. Using the observed judge evaluations and estimated fan votes, we simulated weekly eliminations under different scoring systems and compared the predicted and actual eliminated contestants. We found little difference between the rank and percentage systems, and no clear trend for judge or fan vote favoritism.

Finally, we proposed a new three-phase dynamic scoring system and tested it's outcomes with the same simulation as above. Our system initially weights fan and judge preferences equally to keep fan favorites and allow for growth, before changing the weights to favor judge scores halfway through to prioritize demonstrated dancing ability. We switch back to equal weighting for the finale, to give fans a say in the crowned winner. We found that our system eliminated contentious contestants slightly earlier, in effect balancing the desire of the voters and the integrity of the show by successfully alleviating controversy but not drastically undermining fan favorites.

We have effectively Waltzed, Cha-Cha'd, Tangoed, Jived, and Discoed our way through this comprehensive analysis. The world is marvelous, and so is Dancing with the Stars!

8 Report for the Production Team of Dancing with the Stars

Dancing with the Stars (DWTS) is one of the most iconic cable television shows of the 21st century, though at times it is also the most controversial. The combination of judge evaluations and fan popularity to decide on contestants' outcomes has led to eliminations and even winners that displease or confuse the audience. This contention is exacerbated by the hidden nature of fan votes, while judge scores are aired for each contestant.

After extensive analysis to estimate the fan votes, we offer key insights into how scoring methods impact the eliminations and placements on the show each week, along with our recommendations for a fairer and more engaging competition show.

All 34 seasons of DWTS have equally weighted the judge evaluations and fan popularity, expressed either through rankings or percentages. These mechanisms have proven unfavorable as several controversial contestants have occurred, such as Jerry Rice in season 2 who was the runner up despite receiving the lowest judges scores for several weeks. Similarly, in Season 4 Billy Ray Cyrus placed 5th despite several weeks of last place judge scores. Several show fans found their duration an upset to the dancing integrity of the show.

The show introduced an extended scoring system around Season 28 to mitigate controversy; the bottom two contestants were determined regularly, after which the judges selected who to eliminate. We assert that this system is also suboptimal as it puts too much obvious focus and weight on the judges. It is both poor optics for the judges and unfair to the fans. It is also uninteresting, as most of the time the judges decide to send home the contestant with the lowest judge score anyways.

We propose a new method that removes the conspicuous preference for judge votes, while still allowing for popularity to overpower in the case of breakout fan favorites. It also dynamically adjusts to reflect different aspects of the show being prioritized as the weeks progress.

We use vote percentages as they more precisely reflect both the judges' and public opinion. Equal weighting is used for every week at or before the halfway mark as the fairest way to incorporate both fan and judge preferences. It does not overly penalize contestants unfamiliar with dancing, or unknown to the public. Additionally, it also avoids fan favorites leaving the season early, which would harm viewership and ratings.

For every week after the halfway mark (excluding the finale), we change the weighting to favor the judges scores:

$$\textbf{Score} = 0.6(\textbf{Judge Percentage}) + 0.4(\textbf{Fan Percentage}).$$

At this point in the show, it can become grating for audiences to see poor dancers continue to advance without improvement, as evidenced by the highlighted controversies. Thus, we now prioritize dancing ability to weed out the poorly performing contestants.

For the finale, we revert back to an equal weighting system. Due to the prior weighting scheme, the finale is now likely to only include the better dancers on the show, so we can safely reduce the judge-favored weighting. It is also preferable for fan votes to carry more weight in the finale, as fans value having a say in the final winner.

For a case study, we implemented our method on seasons 2 and 4 using the weekly observed judge scores and estimated fan votes. We found that our method helped mitigate controversy by eliminating the controversial contestant earlier (Table 8).

Table 8: Placement of controversial contestants under equally vs dynamically weighted scoring

Contestant	Season	Equally Weighted Scoring	Dynamically Weighted Scoring
Jerry Rice	Season 2	Runner up	Eliminated week 7
Billy Ray Cyrus	Season 4	Eliminated week 8	Eliminated week 7

Notably, our system eliminates contentious contestants only slightly earlier, which balances the desire of the voters and the integrity of the show by successfully alleviating controversy but not drastically undermining fan favorites.

We hope our report provides you with useful analysis that contributes to making Dancing with the Stars fair and exciting for the contestants, judges, and viewers.

9 References

[1] "“DWTS” Producer Explains How Judges Scores and Audience Votes Are Calculated." EW.com, 2025, https://ew.com/dwts-producer-reveals-how-scores-and-votes-are-calculated-11857082.

[2] Keetharuth, Anju Devianee et al. “Estimating a Preference-Based Index for Mental Health From the Recovering Quality of Life Measure: Valuation of Recovering Quality of Life Utility Index.” Value in health : the journal of the International Society for Pharmacoeconomics and Outcomes Research vol. 24,2 (2021): 281-290. doi:10.1016/j.jval.2020.10.012.

[3] Wu, Xingli, and Huchang Liao. “A Preference Learning Method to Estimate Consumer Preferences from Online Reviews.” Journal of Business Research, vol. 201, 29 Sept. 2025, pp. 115741–115741, https://doi.org/10.1016/j.jbusres.2025.115741.

[4] Bandeen-Roche, Karen, et al. An Introduction to Latent Variable Modeling. 2016.

[5] Heavens, Alan, and Bayesian Models. Bayesian Hierarchical Models ICIC Data Analysis Workshop. 2018.

[6] Terzi, Erol, and Mehmet Ali Cengiz. “Bayesian hierarchical modeling for categorical longitudinal data from sedation measurements.” Computational and mathematical methods in medicine vol. 2013 (2013): 579214. doi:10.1155/2013/579214

[7] Ville. “Chapter 6 Hierarchical Models | Bayesian Inference 2019.” Github.io, 13 Mar. 2019, https://vioshyvo.github.io/Bayesian_inference/hierarchical-models.html.

[8] Zach Seemayer. "“Dancing with the Stars”: Alexis Ren Says Showmance with Alan Bersten Makes Dancing “More Fun” (Exclusive).” Entertainment Tonight, 30 Oct. 2018, https://www.etonline.com/dancing-with-the-stars-alexis-ren-says-showmance-with-alan-bersten-makes-dancing-more-fun-exclusive.